02-750 Week 8

Automation of Scientific Research

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Background: Regression

The goal of regression analysis is to learn a map / function, $f : \mathbb{R}^d \to \mathbb{R}$

- As is the case for learning classifiers, the choice of training instances impacts generalization error, thus, we want to select them actively.

Regression models (like classifiers) can be divided into two categories:

- **Parametric:** the model **summarizes** the training data using a **fixed** number of parameters (independent of the size of the training data)
  - For example, linear regression: $y = x^T\beta$
- **Non-parametric:** the number of parameters is **proportional** to the amount of training data, and the predictions are often made using the actual training instances
  - For example, Gaussian Process Regression, KNN regression, Regression Trees

Background: Bias-Variance Decomposition

A model's **generalization error** can be decomposed as follows:

$$Err(x) = Bias^2 + Variance + Irreducible\ Error$$

- Bias $:= \mathbb{E}_D[\hat{f}(x; D)] - f(x)$
  - This is the error due to the difference between the 'true' function and the one we select from our chosen hypothesis class (Recall PAC learning)
- Variance $:= \mathbb{E}_D\left[\left(\mathbb{E}_D[hat f(x; D)] - \hat{f}(x; D)\right)^2\right]$
  - This is the error due to the training data being a finite sample of $P(X, Y)$. Different training sets will produce different parameters, creating variance.
- Irreducible Error: This is the error due to any noise in the training data

The bias-variance tradeoff

- Bias and variance are generally in opposition to one another
  - Complex models tend to have low bias but high variance
  - Simple models (i.e., those with few parameters) tend to have high bias but low variance
- Some learning algorithms intentionally introduce bias in order to decrease variance
  - For example, **Lasso** and **Ridge** regression introduce bias to reduce variance

Regression Analysis

- The field of Statistics has developed a variety of techniques for learning regression models efficiently (i.e., with low sample complexity)
- Examples:
  - Latin Hypercube Sampling
  - Randomized Block Designs
  - Taguchi Methods
  - Optimal Design of Experiments (DOE)

Optimal Design of Experiments

A branch of statistics devoted to pre-selecting sets of **experimental conditions** that are collectively optimal **with respect to some criterion**

- DOE shares many of the same objectives as Active Learning but...
  - Most DOE techniques are specifically for learning **parametric** regression models
  - Most DOE methods select **all** the experiments *a priori*, not iteratively*
  - **Some DOE methods are more than 100 years old!**

DOE methods often make (strong) assumptions about:

- The **parametric form** of the model (e.g., linear)
- The nature of the noise
- The test density, $P(X)$

Example: DOE for Linear Regression

Model (in matrix form for k training instances): $\mathbf{y} = X\beta + \epsilon$

- $\mathbf{y}$ is a $k \times 1$ vector containing $k \geq 1$ observations / samples
- $\mathbf{X}$ is the $k \times b$ **design matrix**. Here, b is the number of parameters.
- β is the $b \times 1$ parameter vector
- ϵ is a $k \times 1$ noise / error vector
  - Assumptions
    * $\mathbb{E}[\epsilon_i] = 0$. (Expected value of noise is zero)
    * $Var(\epsilon) = \sigma^2 < \infty$ (Noise is constant and has finite variance)
    * $\mathbb{E}[\epsilon_i \epsilon_j] = 0$ (Noise is uncorrelated across instances)

The standard way of estimating β via **ordinary least squares** (OLS)

- Given a vector of $k \geq b$ observations / samples the OLS estimate is:

$$\hat{\beta} = (X'X)^{-1}X'y$$

  - Once we estimate $\hat{\beta}$, the predicted value for any x is: $\hat{y} = x\hat{\beta}$
- It can be shown (via the Gauss-Markov Theorem) that:

- OLS produces a $\hat{\beta}$ that minimizes $\sum_{i=1}^{k}(\hat{y}_i - y_i)^2$
- $\mathbb{E}[\hat{\beta} - \beta_{true}] = 0$ (i.e., OLS is an unbiased estimator)
- $Var(\hat{\beta}) = \sigma^2(X'X)^{-1}$
- $Var(\hat{y}) = \sigma^2 x(X'X)^{-1}x'$

Recall the bias-variance decomposition:

$$Err(x) = Bias^2 + Variance + Irreducible\ Error$$

- OLS is unbiased, so $Err(x) = Variance + Irreducible\ Error$
- We know that $Var(\hat{\beta}) = \sigma^2(X'X)^{-1}$
  - σ^2 is usually unknown, but it is a constant, so we can ignore
  - So the **design matrix**, X is really the **only thing that matters**, not the y's.

DOE methods seek to minimize the variance in the estimate (and thus generalization error) by selecting an optimal set of k training instances (i.e., rows of X)

- Notice that these instances can be selected without knowing **any** labels!
- Different DOE methods use different definitions for what it means to be optimal

Quantifying Uncertainty for Linear Regression

- Given $D_L = \{(x_1, y_1), \cdots, (x_n, y_n)\}$, the sum of squared vertical deviations from the point to the line is

$$f(b_0, b_1) = \sum_{i=1}^{n}[y_i - (b_0 + b_1 x_i)]^2$$

- The point estimates of β_0 and β_1, denoted by $\hat{\beta}_0$ and $\hat{\beta}_1$, are the **least squares estimates**, that is, the values that minimize $f(b_0, b_1)$

$$\begin{aligned}
b_1 = \hat{\beta}_1 &= \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{s_{xy}}{s_{xx}} \\
b_0 = \hat{\beta}_0 &= \frac{\sum_{i=1}^{n} y_i - \hat{\beta}_1 \sum_{i=1}^{n} x_i}{n} = \bar{y} - \hat{\beta}_1 \bar{x} \\
s_{xy} &= \sum_{i=1}^{n} x_i y_i - \frac{\sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n} \\
s_{xx} &= \sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n} \\
\hat{\sigma}^2 = s^2 &= \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{n-2} = \frac{1}{n-2}\sum_{i=1}^{n}\hat{e}_i^2
\end{aligned}$$

Optimality Criteria: G-Optimality

$$\min_{x_i, i=1:k} \max_{x \in \mathcal{X}} \mathrm{Var}(\hat{y}_x)$$

- This criterion selects the k instances that minimize the maximum variance of any prediction
  - This is equivalent to minimizing the maximum element in the diagonal matrix $X(X'X)^{-1}X'$
- This is the criterion proposed by Smith

Optimality Criteria: D-Optimality

$$\max_{x_i, i=1:k} |X'X|$$

- Select the k instances that maximize the *determinant* of $X'X$
- Geometric Interpretation
  - $|X'X|$ is inversely proportional to the product of the axes of the **confidence ellipsoid** around the point estimate of our parameter vector β
  - Bigger $|X'X|$ means smaller ellipsoid, thus, higher confidence in β

Optimality Criteria: A-Optimality and E-Optimality

A-Optimality: (Chernoff 1953)

$$\min_{x_i, i=1:k} \mathrm{Tr}\left((X'X)^{-1}\right)$$

- Select the k instances that minimize the *trace* of the inverse $X'X$
- Minimizes the average variance of the parameter estimates

E-Optimality: (Ehrenfeld 1953)

- Select the k instances that maximizes the minimum eigenvalue of $X'X$
- Minimizes the maximum variance of β

DOE for Nonlinear Models

If the "true" model is non-linear, but can be expressed as a linear **basis function**, we can still use the previous objectives to find optimal designs

- A linear basis function has the form: $f(x) = \sum_{i=1}^{b} \alpha_i \phi_i(x)$
  - Here, each ϕ_i is one of the basis functions
- Common Basis Functions: Polynomial, Fourier, Gaussian, Sigmoid

Example:

Consider the polynomial $f(x) = \alpha_1 c + \alpha_2 x + \alpha_3 x^3 + \alpha_4 x^3$

- The basis functions are $\phi_1 = c, \phi_2 = x, \phi_3 = x^3, \phi_4 = x^3$
- We could construct a $k \times 4$ design matrix, where the points are selected using any of the previously defined criteria (e.g., G-optimality)

$$X = \begin{bmatrix} 1 & x_1 & x_1^2 & x_1^3 \\ 1 & x_2 & x_2^2 & x_2^3 \\ 1 & x_3 & x_3^2 & x_3^3 \\ 1 & x_4 & x_4^2 & x_4^3 \end{bmatrix}$$

- After obtaining labels, we can use OLS to estimate $\alpha = [\alpha_1 \alpha_2 \alpha_3 \alpha_4]'$

Summary: Optimal Design of Experiments

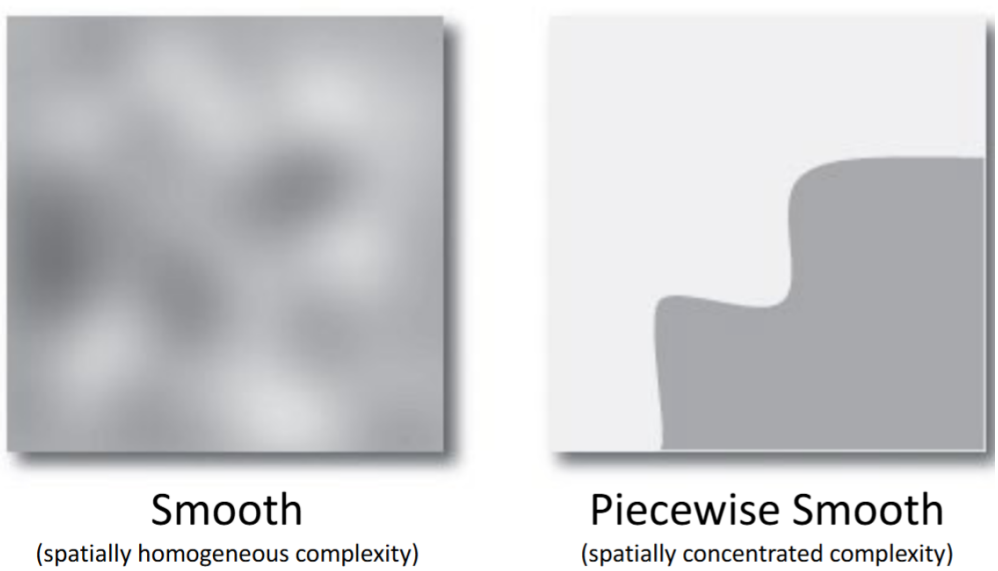
- DOE is primarily used for learning parametric regression models that can be expressed as linear basis functions
- DOE is similar to Active Learning in the sense that it selects training data
  - DOE methods select instances by optimizing some scalar property of $X'X$ which will, in turn, minimized the (expected) generalization error
  - Unlike active learning, DOE methods select the design points at the same time

Recursive Dyadic Partition (RDP) Algorithm

- RDP is an algorithm for learning **non-parametric regression models**
- Type II (cluster expoiltation). However, unlike ZLG, DH, and PLAL, it is intended for the **membership query synthesis** data access model.
  - Pool-based sampling is possible, if certain conditions are met
- It is **aggressive**
- Guarantees:
  - RDP will converge to the optimal model
  - The upper and lower bounds on the rate of convergence are known

Rationale

RDP is intended for learning **piecewise smooth functions**.



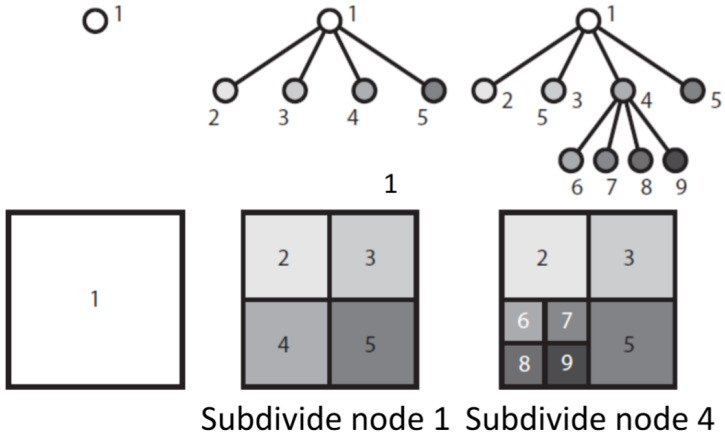
For functions whose complexity is spatially non-uniform and highly concentrated in small subsets of the domain, **the extra spatial adaptivity of the active learning paradigm can lead into significant performance gains**

- The authors show that **there is no benefit to active learning** (relative to passive learning), if the function satisfies the **Hölder condition**.

RDP: Dyadic Space-Partitioning Trees

The algorithm uses dyadic space-partitioning trees.

- Example in a 2D domain (2^d-ary tree structure)



- Consider $f : [0,1]^d \to \mathbb{R}$
  - Step 1: Divide $[0,1]^d$ into 2^d equal sized hypercubes
  - Step 2: Repeat this process again on each hypercube
- Repeating this process $\log(m)$ times gives rise to a partition of the unit cube into m^d hypercubes of identical size. (This process is represented as a 2^d-ary tree structure above.)

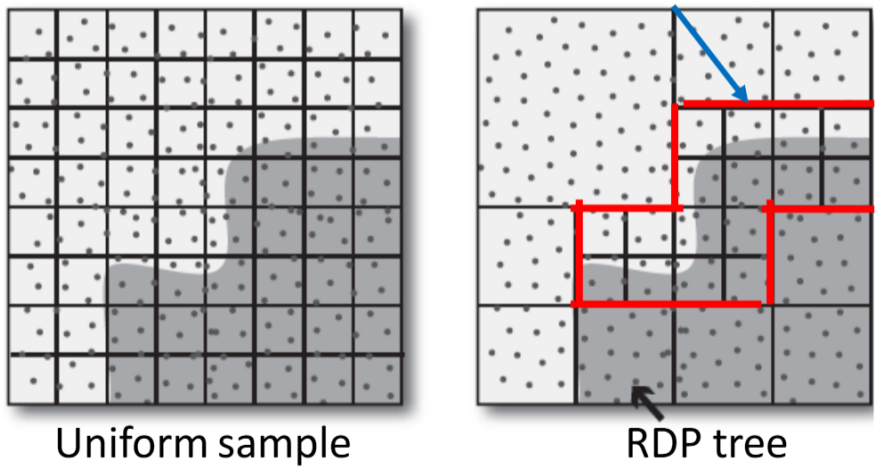
Note: Dyadic means to divide the domain by 2^d, where d is the dimension of $\mathcal{X} = [0,1]^d$

There are **two phases** for this algorithm.

1. **Preview Phase:**
   - Use half of the labeling budget to label points **uniformly** within the domain, $\mathcal{X}$
   - Learn a bounded height regression tree modified to partition the domain via a **Recursive Dyadic Partition** (RDP)
   - Use the RDP model to identify boundaries in the domain
2. **Refinement Phase:**
   - Use the remaining half of the budget to label the boundary regions densely
   - Learn a regression model from the labeled points

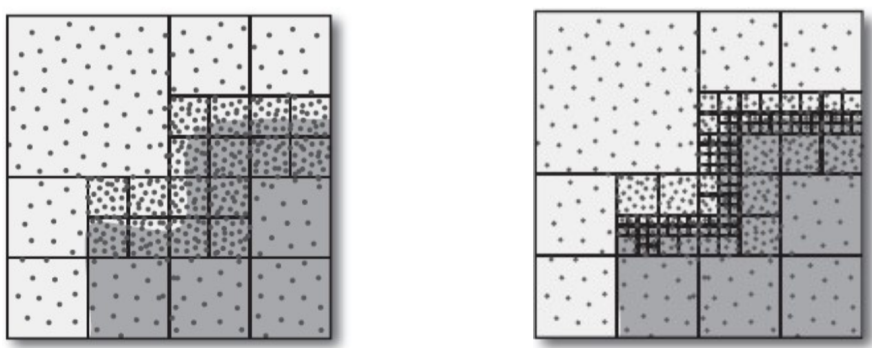
Example:

Preview Phase:



The deepest nodes in the tree will tend to be in boundary regions (i.e., places where the function changes quickly)

Refinement Phase:



Use the remaining budget to desnsely sample the regions of the domain corresponding to the deepest nodes of the regression tree

Technical Issue

- If the dyadic partition is aligned with a boundary, the refinement phase is unlikely to sample in that region
- This can be addressed by building multiple RDP models, shifted along one or more dimensions. For example, the bottom left corner in the above examples.

RDP Guarantees

Consistency and Rate of Convergence (under certain assumptions on f)

- The authors derive upper and lower bounds for the rate of convergence to the optimal model
  - Upper bound: $\mathbb{E}\left[\|\hat{f} - f\|^2\right] \leq C\left(\frac{n}{\log n}\right)^{\frac{1}{d-1+\frac{1}{d}}}$
  - Lower bound: $\mathbb{E}\left[\|\hat{f} - f\|^2\right] \leq cn^{-\frac{1}{d-1}}$
- The corresponding bounds for passive learning are as follows:
  - Upper bound: $\mathbb{E}\left[\|\hat{f} - f\|^2\right] \leq C\left(\frac{n}{\log n}\right)^{-\frac{1}{d}}$
  - Lower bound: $\mathbb{E}\left[\|\hat{f} - f\|^2\right] \leq cn^{-\frac{1}{d}}$

- That is, RDP's complexity is essentially that of passive learning, but on a lower-dimensional space

RDP Algorithm: Summary

- The algorithm is elegant and easy to implement, but **the number of samples needed to obtain reasonable accuracies will be large in practice**, since we need to guarantee sufficient uniform sampling in a d-dimensional space, which is expensive (see *curse of dimensionality*)
- If desired, we can use the labeled examples to train a more sophisticated regression model
- The algorithm is noteworthy because it is **aggressive**, yet the authors are able to prove theorems about learning rates
  - Note, however, that these guarantees only hold if the underlying function is piecewise continuous

Summary: DOE vs. Active Learning

Objective: Generate dataset from which one can learn an accurate predictive model

Category	Design of Experiments	Active Learning
Logistics	Limited to one batch (e.g., clinical trials)	Multiple batches (e.g., compound screening)
Feature knowledge	Good when you know all your features are useful	Good when you don't know if your features are useful
Model assumptions	Typically assumes data are derived from a specific model	No assumptions needed

ALICE

ALICE: "Active Learning in Approximately Linear Regression Based on Conditional Expectation of Generalization Error", Sugiyama (JMLR, 2006)

Aleatoric Uncertainty vs. Epistemic Uncertainty

- **Aleatoric uncertainty:** also known as data uncertainty, is irreducible uncertainty in data that gives rise to uncertainty in predictions
  - This type of uncertainty is not a property of the model, but rather is an inherent property of the data distribution, and hence, it is irreducible
- **Epistemic uncertainty:** also known as knowledge uncertainty, occurs due to inadequate knowledge
  - One can define models to answer different questions in model-based prediction

The ALICE Algorithm

Key Features:

- ALICE is an algorithm for learning parametric regression models that take the form of **linear basis functions**
- Unlike most query selection methods, **it is not iterative!**
  - It constructs an A-optimal design, and then queries those points all at once
  - It is intended for the membership query synthesis data access model
    - Pool-based sampling is possible, if certain conditions are met

- It is **aggressive**, but corrects for sampling bias using **importance weighting** (similar to IWAL)
- Guarantees
  - ALICE will converge to the optimal model
  - Under the assumption that the true model is approximately linear, the author proves bounds on the learning rate

ALICE: Motivation

Traditional DOE methods **assume that we know the form of the true function**

- Suppose that we are wrong. How bad could the generalization error be?
- For example:
  - True function: $f(x) = 1 - x + x^2 + \delta x^3 + \epsilon$
  - Assume that $\epsilon \sim \mathcal{N}(0, 0.3)$
  - Our assumed model: $g(x) = \alpha_1 - \alpha_2 x + \alpha_3 x^2$
  - Compare A-optimal design vs a random selection of $k = 10$ labeled instances

Generalization Error (k = 10)	$\delta = 0$	$\delta = 0.05$	$\delta = 0.5$
A-optimal design: $\min \mathrm{Tr}\left((X'X)^{-1}\right)$	1.1	2.0	95.5
Random Selection	2.2	2.3	7.1

- The magnitude of the error for $\delta = 0.5$ is due in part to the bias we introduce by misspecifying the model

ALICE: Background

It can be shown that for any linear estimator that:

$$Variance = \sigma^2 \mathrm{Tr}(ULL') \propto \mathrm{Tr}(ULL')$$

where $L = (X'X)^{-1}X'$ is the **Moore-Penrose pesudoinverse** and

$$U_{i,j} = \int \phi_i(x)\phi_i(x)P_{\text{test}}(x)\,\mathrm{d}x$$

is a $b \times b$ matrix, where b is the number of terms in the linear basis function

- Recall that **for a correctly specified model**, ordinary least squares (OLS) is an unbiased estimator, thus, **minimizing Tr**(ULL') **will minimize the (expected) generalization error**.

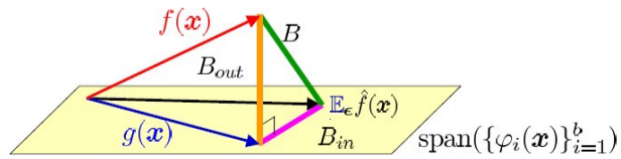
The author decomposes bias into two orthogonal components:

- The **out-of-model bias** due to misspecifying the model;

$$B_{out} = \int \left(f(x) - g(x)\right)^2 P_{\text{test}}(x)\,\mathrm{d}x$$

- The **in-model bias** due to differences in P_{train} and P_{test}

$$B_{in} = \int \left(\mathbb{E}_{\text{train}}[\hat{f}(x)] - g(x)\right)^2 P_{\text{test}(x)\,\mathrm{d}x}$$



- If we have misspecified the model, $B_{out} > 0$
  - But, we can't reduce it either (unless we are **told** the true form of the function)
    - That is, B_{out} is a kind of irreducible error
  - So, we can just ignore B_{out} when it comes to selecting design points
- OLS is an unbiased estimator, so is it reasonable to assume that $B_{in} = 0$?
  - Unfortunately, OLS is only unbiased if $P_{train} = P_{test}$, which isn't true in Active Learning, so $B_{in} > 0$
  - Solution: Use **Importance Weighting** (just like the IWAL algorithm)

ALICE's overall strategy is to use importance weighting to reduce B_{in}, and then to use an A-optimal design criterion to reduce variance, and thus the generalization error

Recall: Importance Weighting

- The empirical average loss for model h on n instances is: $\frac{1}{n}\sum_i loss(h(x_i), y_i)$
- As n goes to ∞ the training loss converges to

$$\int loss(h(x), y) P_{train}(x)\, \mathrm{d}x$$

- Of course, what we really care about is the **test** loss:

$$\int loss(h(x), y) P_{test}(x)\, \mathrm{d}x$$

  - If $P_{train} = P_{test}$, then these losses will be the same
  - But if we have a biased training data, then $P_{train}(\mathcal{X}) \neq P_{test}(\mathcal{X})$
- Recall that many learning algorithms can use **weighted loss functions**
  - Ex. weighted empirical loss: $\widehat{err}(h) = \frac{1}{n}\sum_i w_i loss(h(x_i), y_i)$
- Let the **importance weight** of a training instance be defined as:

$$w_i = \frac{P_{test}(x_i)}{P_{train}(x_i)}$$

- The importance-weighted loss is

$$\widehat{err}_{IW}(h) = \frac{1}{n}\sum_i \frac{P_{test}(x_i)}{P_{train}(x_i)} loss(h(x_i), y_i)$$

- As n goes to ∞ the importance-weighted loss converges to:

$$\begin{aligned}\widehat{err}_{IW}(h) &= \int loss(h(x), y) \frac{P_{test}(x)}{P_{train}(x)} P_{train}(x)\, \mathrm{d}x \\ &= \int loss(h(x), y) P_{test}(x)\, \mathrm{d}x\end{aligned}$$

  which is exactly what we want!
- Thus, if we can compute suitable importance weights, the fact that our training data are biased will not matter (in the limit)

ALICE and Importance Weighting

- Recall the **oridinary least squares** (OLS) estimate for a linear model $y = X\beta$ is:

$$\hat{\beta} = (X'X)^{-1}X'y$$

- The **importance-weighted Least Squares** estimate minimizes the weighted sum of errors

$$\hat{\beta} = (X'DX)^{-1}X'Dy$$

  where

$$D_{i,j} = \begin{cases} \frac{P_{test}(x_i)}{P_{train}(x_i)} & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}$$

Properties of **Importance-weighted Least Squares**

- It is unbiased, asymptotically, even if the model is misspecified!
- If the model is approximately correct (i.e., $f(x) \approx g(x)$), then the in-model bias is much smaller than the variance (i.e., $B_{in} <<$ Variance)
- If we re-define matrix $L \equiv (X'DX)^{-1}X'D$, then the variance is still

$$Variance = \sigma^2 \mathrm{Tr}(ULL') \propto \mathrm{Tr}(ULL')$$

  - So, we can still use an A-optimal design criterion to reduce variance, and thus the generalization error

ALICE: Implementation Details

- The traditional approach to finding A-optimal (or D-optimal, etc.) designs is to use **semi-definite programming**
- ALICE doesn't use semi-definite programming
  - Instead, it takes as input a finite set of k probability densities, $\hat{P} = \{P_1, \cdots, P_k\}$
  - It **samples** n instances from each $p \in \hat{P}$ and then
    - Constructs k design matrices, $X_1, \cdots, X_k$, one for each $p \in \hat{P}$
    - Constructs the corresponding $L_1, \cdots, L_k$ matrices
  - The algorithm finds the matrix L_i that minimizes $\mathrm{Tr}(UL_iL_i')$ and then requests for the corresponding set of points
  - It then learns the regression model using importance-weighted least squares

The ALICE Algorithm

- **Input:** A finite set $\hat{\mathcal{P}}$ of strictly positve probability densities
- Calculate U.

$$U_{i,j} = \int \phi_i(x)\phi_j(x)P_{test}(x)\,\mathrm{d}x$$

- For each $p \in \hat{\mathcal{P}}$
  - Create training input points $\{x_i^{(p)}\}_{i=1}^n$ following $p(x)$
  - Calculate L_W

$$L = (X'DX)^{-1}X'D$$

- Calculate $J(p)$

$$J(p) = \mathrm{Tr}(ULL')$$

- Choose $\hat{p}$ that minimizes J
- Put $x_i = x_i^{(\hat{p})}$ for $i = 1, 2, \ldots, n$.
- Call the oracle and learn the model $\alpha = Ly$, which entails:
  - Observe the training output values $\{y_i\}_{i=1}^n$ at $\{x_i\}_{i=1}^n$
  - Calculate $\hat{\alpha}_W$ by Eq. (9)
- **Output:** $\hat{\alpha}_W$

ALICE Guarantees

Error (assuming that $f(x) = g(x) + \delta r(x)$)

- $\mathbb{E}[Err(x)] = \sigma^2 \mathrm{Tr}(ULL') + O(\delta n^{-1})$
  - Note that $\sigma^2 \mathrm{Tr}(ULL') = O(n^{-1})$, so if $\delta = O(1)$, then the expeted error decreases as $O(n^{-1})$ which is the same asymptotic rate as if the model was specified correctly

Example Results

- True function: $f(x) = 1 - x + x^2 + \delta x^3 + \epsilon$
  - Assume that $\epsilon \sim \mathcal{N}(0, 0.3)$
- Our assumed model: $g(x) = \alpha_1 - \alpha_2 x + \alpha_3 x^2$
- Compare three different strategies using $k = 10$ labeled instances

Generalization Error (k = 10)	$\delta = 0$	$\delta = 0.05$	$\delta = 0.5$
ALICE	1.1	1.2	4.7
A-optimal design: $\min \mathrm{Tr}\left((X'X)^{-1}\right)$	1.1	2.0	95.5
Random Selection	2.2	2.3	7.1

ALICE: Summary

- ALICE actively learns **linear basis function models**, and provides generalization guarantees, **even if the model is misspecified**
- It is intended for a membership query synthesis data access model
- It uses two techniques to (approximately) minimize generalization error
  - **Importance weighting** corrects for in-model bias, resulting in an unbiased estimator
  - The importance weighted **A-optimal design** reduces variance
- It has the same learning rate as passive learning, but produces better models
- **It assumes that we know** P_{test}
  - If you don't know P_{test} but do have access to an unlabeled pool of samples from P_{test}, the same author does have a closely related algorithm:
    - Pool-based active learning in approximate linear regression

LapRDD

"Laplacian Regularized D-optimal Design for Active Learning and its Application to Image Retrieval", He (IEEE Trans Image Process, 2009)